

Semester 1 Week 6 LAB SIFT

Task 1 (Image_Stitching.ipynb)

Open the image TowerFull.jpg using an image editor (Paint).

Select the left side about 2/3 of image, crop it and save it as TowerRight.jpg

Re-Open the image TowerFull.jpg using an image editor (Paint).

Select the right side about 2/3 of image, crop it and save it as TowerLeft.jpg

No need to resize images, TowerRight.jpg & TowerLeft.jpg, because vertically they are of same dimensions.

Upload these images in Jupyter-Lab or in the current folder.

In the code (Image_Stitching.ipynb), ensure proper naming of Left and Right Images

```
image1 = cv2.imread('TowerLeft.jpg')
```

```
image2 = cv2.imread('TowerRight.jpg')
```

Run/Execute the program, view output results.

This is the easiest and most efficient way to test image stitching using SIFT, because we have generated the left and right images manually from an original source image.

Repeat the above and test with two other sample images: CafeteriaFull.jpg & AgriBuildingFull.jpg

Task 2 (Image_Stitching.ipynb)

Test and run the program again using another pair of images (ZurichImageA.jpg & ZurichImageB.jpg)

Determine which image (ZurichImageA.jpg & ZurichImageB.jpg) is Left view and Right view for proper stitching.

Check the dimensions of the two images m x n pixels.

Upload these images in Jupyter-Lab or in the current folder.

In the code (Image_Stitching.ipynb), ensure proper naming of Left and Right Images

```
image1 = cv2.imread('ImageNameleft.jpg')
```

```
image2 = cv2.imread('ImageNameright.jpg')
```

Run/Execute the program, view output results.

Use a camera take two pictures of a building on campus, 1st picture capturing leftward view of building and 2nd picture the rightward view of the building. The two pictures must be taken with just a slight shift (panning) of the camera.

Resize the two pictures if needed.

Repeat the image stitching experiment as previously done. What difficulties and issues did you encounter?

Task 3 (SIFT_Matching.ipynb)

Upload your pair of images (Leftpart & Rightpart) in Jupyter-Lab or in the current folder.

In the code (SIFT_Matching.ipynb), ensure proper naming of Left and Right Images

```
train_img = cv2.imread('Leftpart.jpg')
```

```
query_img = cv2.imread('Rightpart.jpg')
```

Run/Execute the program, view output results and comment on the detected keypoints.

Task 4 (SIFTImage_Match.ipynb)

In the code (SIFTImage_Match.ipynb), ensure proper naming of Left and Right Images.

Run/Execute the program, view output results and comment on the detected keypoints.

Number of keypoints detected?